

Lab Session Task 1

We assume the following for a function $f(x)$ with a single input x :

- (i) The mathematical function form $f(x)$ is not available.
- (ii) The value of $f(x)$ can be calculated for any input value.
- (iii) The optimal solution is known to be in the interval $[0, 1]$
- (iv) $f(x)$ is unimodal in $[0, 1]$ for minimization.

Q1. When we can examine only two solutions, how we can specify those solutions to minimize the interval including the optimal solution?

Q2. When we can examine only three solutions, how we can specify those solutions?

Q3. When we can examine 20 solutions, how we can specify those solutions ?

Q4. In addition to (i)-(iv), we also assume that the value of $f'(x)$ can be calculated for any input value. When we can examine 20 solutions, how we can specify those solutions ?

Q5. Create a function $f(x)$ in $[0, 1]$ on which Newton's Method does not work well on some initial solutions.

Lab Session Task 2

For each of the following problems, (i) mathematically calculate the solution, and (ii) numerically estimate the solution through computer simulations with a large number of trials.

- (1) We randomly generate two points A and B in $[0, 1]^2$. Calculate the probability of A being dominated by B .
- (2) We randomly generate two points A and B in $[0, 1]^4$. Calculate the probability of A being dominated by B .
- (3) We randomly generate two points A and B in $[0, 1]^{10}$. Calculate the probability of A being dominated by B .
- (4) We randomly generate 200 points in $[0, 1]^2$. Calculate the expected number of non-dominated solutions among them.
- (5) We randomly generate 2000 points in $[0, 1]^2$. Calculate the expected number of non-dominated solutions among them.
- (6) We randomly generate 200 points in $[0, 1]^{10}$. Calculate the expected number of non-dominated solutions among them.
- (7) We randomly generate 2000 points in $[0, 1]^{10}$. Calculate the expected number of non-dominated solutions among them.